

## 1 Question (15 points)

Consider a network that is a rooted tree, with the root its source, the leaves as its sinks, and all the edges directed along the paths from the root to the leaves. Design an efficient algorithm for finding a maximum flow in such a network.

## 2 Question (15 points)

A group of college courses and a group of classrooms are given. Based on its enrollment and audio-visual needs, not all courses can be taught in all classrooms. You are given a list of courses and classrooms, where course  $X$  can be taught in classroom  $Y$ . Your task is to find the largest set of college courses that can be taught simultaneously without conflicts.

### 3 Question (15 points)

You are given a 3-Sat problem with  $k$  variables. Find if there is an assignment of variables to the problem such that CNF is satisfied using backtracking.

## 4 Question (15 points)

You are given a weighted graph  $G$ . Find the longest path in graph  $G$  using backtracking.